

OCT Research Landscape: Disease Burden & Publishable Directions by Modality

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This maps where the global eye-disease burden concentrates, what each generation of optical coherence tomography can image, and the research directions — across spectral-domain OCT (SD-OCT), swept-source OCT (SS-OCT), and OCT angiography (OCT-A) — that are currently open enough to support a strong, publishable contribution. Directions are framed as research opportunities the field presents, with the disease burden and dataset landscape that motivate them.

Global disease burden: where the need concentrates

Three conditions dominate the worldwide burden of vision-threatening retinal disease, and all three are where OCT-based biomarkers are clinically decisive. This is what motivates the research directions that follow.

Disease	Current global burden	Projection	Why it matters for OCT research
Diabetic retinopathy (DR) / diabetic macular edema (DME)	~103M with DR (2020/21); DME/CSME ~19M; ~22% of people with diabetes have DR	DR ~161M by 2045 (diabetes 537M → 783M)	Leading cause of vision loss in working-age adults; fluid and capillary changes are directly measurable on OCT and OCT-A.
Age-related macular degeneration (AMD)	~196M (2020), ~8.7% of the population; ~18.6M with advanced AMD; accounts for ~6–9% of global blindness	~288M by 2040	Drusen, fluid, and geographic atrophy are OCT-defined; newly approved GA therapies make automated OCT quantification urgent.

Disease	Current global burden	Projection	Why it matters for OCT research
Glaucoma	~76M (2020), the leading cause of irreversible blindness	~112M by 2040	Retinal nerve fibre layer and ganglion-cell thinning on OCT support early detection and progression tracking.

Reading the table. DR/DME and AMD are the two giants and the natural disease anchors for OCT, SS-OCT, and OCT-A work; glaucoma is the third pillar. The burden is rising fastest in low- and middle-income regions, where access to eye care is thinnest — which raises the stakes for automated, generalizable tools rather than models tuned to one clinic.

The three OCT generations at a glance

Each generation images different anatomy and sits at a very different level of research saturation — which is what determines where the open problems are.

Generation	Principle	What it images best	Public data	Research saturation
SD-OCT (spectral-domain, ~840 nm)	Spectrometer + line camera; the clinical workhorse	Retinal layers, fluid, drusen, nerve fibre layer	Abundant (Kermany, RETOUCH, Duke, OCTDL, etc.)	High — cross-sectional classification/segmentation near-saturated
SS-OCT (swept-source, ~1050 nm)	Tunable laser; deeper penetration, faster, less roll-off	Choroid, widefield/peripheral retina, through media opacities, anterior segment	Scarce (mostly private, small cohorts)	Low — new anatomy, largely open
OCT-A (angiography; SD- or SS-based)	Motion contrast across repeated scans; images blood flow without dye	Retinal/choroidal microvasculature, foveal avascular zone (FAZ)	Very scarce (ROSE ~229 images, OCTA-500)	Low-medium — vessel/FAZ tasks under-studied

The strategic pattern. SD-OCT is mature and crowded, so entering on its classic tasks means competing on marginal gains. SS-OCT and OCT-A open fresher anatomy but are data-scarce — which pushes the novelty toward label-efficient, self-supervised, generative, and uncertainty-aware methods rather than brute-force supervised training.

Research directions — SD-OCT

SD-OCT's cross-sectional tasks (disease classification, layer and fluid segmentation) are close to solved on curated data, and reported accuracy is often inflated by train/test leakage. The live frontier is longitudinal and prognostic rather than single-scan.

- **Treatment-response prediction (anti-VEGF).** Predicting responders vs non-responders to anti-VEGF therapy from baseline OCT is active but immature — mostly small single-centre studies with moderate performance (recent models around AUC 0.84). Reviews flag two gaps: deep learning is under-used for treatment-effectiveness prediction, and public longitudinal OCT datasets are scarce.
- **Geographic atrophy (GA) quantification and progression.** With complement-inhibitor therapies now approved (pegcetacoplan, avacincaptad), automated GA area and growth-rate quantification became clinically urgent for trial endpoints and monitoring. Current segmentation sits around Dice 0.82–0.83; the open problems are multi-centre/multi-device robustness and predicting future GA growth from 3D volumes, not just measuring current area.
- **Multimodal and interpretable prognosis.** Combining OCT with clinical/blood biomarkers or ultra-widefield imaging, with explicit interpretability (e.g. SHAP, structure-function correlation), to move beyond black-box classifiers.
- **Cross-vendor generalization.** Models degrade across devices (Cirrus, Spectralis, Topcon, etc.); domain-adaptation and vendor-invariant representations remain the biggest barrier to deployment.
- **Foundation models and self-supervision.** Label-efficient pretraining (masked autoencoding, contrastive learning) that transfers across tasks; evaluation frameworks that capture clinical safety, not just accuracy, are still underdeveloped.
- **Uncertainty and reliability.** Calibrated confidence, out-of-distribution detection, and selective prediction (abstaining on unreliable scans) — safety-critical and comparatively immature.

Disease anchors: AMD (incl. GA) and DR/DME.

Research directions — SS-OCT

SS-OCT is where the anatomy is fresh. Longer wavelength penetrates deeper and resists scattering, exposing structures SD-OCT largely cannot resolve — but the modality is confined to specialized centres by cost, so public data is thin. That scarcity is the defining constraint and, in turn, the reason label-efficient methods matter here.

- **Choroidal biomarkers and segmentation.** SS-OCT enabled quantitative choroidal imaging for the first time: choroidal layer segmentation, 3D choroidal vasculature, and choroidal thickness. Machine learning here is early and validated mostly on small cohorts (tens to low hundreds of eyes), leaving room in pachychoroid disease, central serous chorioretinopathy (CSC), and the choroidal component of AMD.
- **Widefield / ultra-widefield analysis.** Faster swept scanning gives 80°+ fields, enabling automated non-perfusion-area quantification on ultra-widefield SS-OCT angiography for DR severity, plus peripheral-lesion detection — an area still being built out.
- **Imaging through media opacities and combined vitreous-choroid scans,** and deeper anterior-segment imaging (angle assessment, cornea) — largely untouched by deep learning.
- **Working within data scarcity.** Because annotated SS-OCT data is limited, self-supervised pretraining, weak/scribble supervision, and synthetic augmentation are the practical routes to competitive performance.

Disease anchors: pachychoroid/CSC, AMD (choroidal changes), high myopia, peripheral DR.

Research directions — OCT-A

OCT-A images blood flow rather than anatomy, making it structurally richer, and it is less saturated than SD-OCT classification. Its constraint mirrors SS-OCT: almost no large public datasets (the main annotated vessel set, ROSE, has ~229 images; OCTA-500 adds a few hundred subjects), and vessel segmentation is under-studied due to low capillary visibility, high vessel complexity, artifacts, and low signal-to-noise ratio.

- **Trustworthy vessel and FAZ quantification.** Vessel density, FAZ area/perimeter, tortuosity, and fractal dimension are reported widely, but reliability is rarely quantified. Calibrated uncertainty, artifact/quality-aware prediction, and abstention on low-quality scans are open — and a quality-labelled set (OCTA-25K-IQA-SEG) exists to validate against.

- **Cross-device generalization.** OCT-A adds vendor variability plus projection differences (superficial vs deep vascular plexus); a benchmark and domain-adaptation method that holds across devices is a clear gap.
- **Oculomics with rigor.** The retina is the only place to non-invasively image microvasculature, and OCT-A biomarkers are being pushed to predict cardiovascular, cerebrovascular, renal, metabolic, and neurodegenerative disease. The field is hot but plagued by overclaiming, shortcut learning (models predicting confounders like age/sex), and thin external validation — so a study with genuine multi-cohort validation, confounder ablation, and calibrated risk is itself the contribution.
- **Self-supervised / foundation model for OCT-A.** Retinal foundation models exist for OCT and fundus (RETFound, MIRAGE, OCTCube) but essentially none is dedicated to OCT-A, despite abundant unlabelled clinical OCT-A — a genuine white space, ideally paired with a multi-task benchmark.

Disease anchors: DR (capillary dropout, FAZ enlargement), AMD (neovascularization), glaucoma (peripapillary perfusion), plus systemic disease via oculomics.

Cross-generation directions

The strongest opportunities often span modalities, using the abundance of one generation to compensate for the scarcity of another.

- **Generative SD→SS choroidal enhancement, with a fidelity audit.** Because SS-OCT is scarce but SD-OCT is everywhere, generative models are being used to synthesize SS-quality choroidal detail from cheap SD-OCT scans (demonstrated in a 2024 npj Digital Medicine study). The open and largely unexamined question is whether the synthesized choroid preserves the biomarkers a clinician would measure, or fabricates/erases vessels — a clean claim-and-validation problem: does enhancement distort the measurement?
- **Cross-generation domain generalization + uncertainty.** SD↔SS is a harder, more physically meaningful domain shift than vendor-to-vendor (different wavelength, penetration, noise) and is barely studied as a generalization problem. A model — or a calibrated uncertainty layer — that flags when an SD-trained model should not be trusted on SS input (or vice versa), validated across generations, addresses a real deployment risk.

- **Trustworthy longitudinal prognosis.** GA growth prediction and anti-VEGF response both suffer from thin longitudinal data, making them ideal settings for uncertainty-aware, label-efficient methods that report calibrated progression risk and abstain when the baseline scan is uninformative.

The recurring practical constraint across all three is data access: SS-OCT and longitudinal SD-OCT realistically require a clinical data partnership to support external-validation claims, so that should be designed in from the start.

What makes a contribution publishable at a top venue

Across all of the directions above, what separates a top-tier contribution from an incremental one is rarely architecture novelty — it is the rigor of the claim and its validation.

- **External / cross-device validation**, not just a better number on one dataset. Evaluation frameworks for these models remain underdeveloped and standard metrics poorly capture clinical safety or utility, so evaluating reliability well is itself a contribution.
- **Confounder and shortcut auditing** — demonstrating that the signal comes from the intended structure (e.g. the vasculature) rather than from age, sex, device, or acquisition artifacts.
- **Claims scoped to the evidence.** Reviewers reward an honest map of where a method breaks far more than an inflated headline number.
- **Attention to data integrity** — patient-level train/test splits and checks for leakage, given that classic benchmarks (e.g. Kermany) have documented overlap that inflates accuracy.

Target venues. Methods framing: MICCAI (and the OMIA ophthalmic-imaging workshop), MIDL, ISBI. Clinical-validation and oculomics framing: Ophthalmology Science, Translational Vision Science & Technology (TVST), npj Digital Medicine, Medical Image Analysis.

Key public datasets by generation

Data availability is one of the strongest predictors of where a direction is tractable. Note how it thins out moving from SD-OCT to SS-OCT and OCT-A.

Dataset	Generation	Task	Scale	Notes
Kermany / OCT2017	SD-OCT	Classification (CNV, DME, drusen, normal)	~84k B-scans	Default benchmark; near-saturated; documented train/test leakage.
OCT-C8 / OCT-8C	SD-OCT	8-class classification	~24k images	Harder multi-class successor.
OCTDL	SD-OCT	7-class classification	~2k+ images	Newer, more disease diversity.
RETOUCH	SD-OCT	Fluid segmentation (IRF/SRF/PED)	70 volumes, 3 vendors	Canonical fluid benchmark (MICCAI 2017).
Duke SD-OCT	SD-OCT	Classification + layer segmentation	Multiple sets	Long-standing layer-segmentation references.
OLIVES	SD-OCT	Multimodal (labels + biomarkers)	~78k OCT	Pairs OCT with clinical/biomarker data.
ROSE	OCT-A	Vessel segmentation	229 images	First public annotated OCT-A vessel set.
OCTA-500	OCT-A	Vessel / FAZ segmentation, classification	~500 subjects	Multimodal, multi-label; widely used.
OCTA-25K-IQA-SEG	OCT-A	Quality assessment + segmentation	~25k images	Large; includes image-quality labels.
SS-OCT choroid sets	SS-OCT	Choroid layer / vasculature	Tens-hundreds of eyes	Mostly private; small cohorts.

Longitudinal (multi-visit) OCT datasets remain especially scarce across all generations — a key bottleneck for prognosis and treatment-response research.

Sources

Key references behind the burden figures, modality comparisons, and research directions (as of 22 Sep 2026):

- [Global prevalence of AMD and projection for 2020 and 2040 \(Wong et al., *Lancet Global Health*\)](#)
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- [The diabetic retinopathy “pandemic” — 2023 Friedenwald Lecture \(IOVS/ARVO\)](#)
- [Swept-source and multimodal OCT clinical advantages \(*Optometry Times*\)](#)
- [Widefield OCTA: advancing retinal imaging \(*Retinal Physician*\)](#)
- [Deep learning classification of choroidal layers in SS-OCT \(*BMC Ophthalmology* 2025\)](#)
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- [Deep learning segmentation of geographic atrophy — multi-centre, multi-device validation \(*Diagnostics* 2025\)](#)
- [Deep learning to predict GA progression using 3D OCT \(2025\)](#)
- [Evaluating anti-VEGF responses in DME — AI review \(*Indian J Ophthalmology* 2025\)](#)
- [OCTA-500 dataset \(*Medical Image Analysis*\)](#)
- [ROSE: retinal OCT-A vessel segmentation dataset](#)
- [AI oculomics for systemic health: 2025 and beyond \(review\)](#)
- [RETFound — foundation model for retinal images \(*Nature* 2023\)](#)
- [Inflation of test accuracy due to data leakage in OCT classification](#)